

3.Score calculator app

Aim:

To Create a react app for Student Management Portal named scorecalculatorapp and create a function component named “CalculateScore” which will accept Name, School, Total and goal in order to calculate the average score of a student and display the same.

Steps:

1.create react app

```
PS C:\Users\laksh> npx create-react-app scorecalculatorapp
Created git commit.

Success! Created scorecalculatorapp at C:\Users\laksh\scorecalculatorapp
app
Inside that directory, you can run several commands:

  npm start
    Starts the development server.

  npm run build
    Bundles the app into static files for production.

  npm test
    Starts the test runner.
```

Codes:

Calculatescore.js:

The screenshot shows a VS Code editor with a JavaScript file named `CalculateScore.js`. The file contains a function `CalculateScore(props)` that calculates an average score and returns an HTML string. The HTML string includes a table with student details: Name, School, Total Marks, and Subjects. The average score is calculated as `props.Total / props.goal`. The table is rendered with Bootstrap classes. The VS Code interface shows the Explorer sidebar with a file tree, the Run and Debug sidebar, and the Output sidebar. The bottom status bar shows the file is at line 10, column 32, and the encoding is UTF-8.

```
function CalculateScore(props) {
  const average = props.Total / props.goal;

  return (
    <div className="box">
      <h2>Student Details</h2>

      <table>
        <tbody>
          <tr>
            <td>Name</td>
            <td>{props.Name}</td>
          </tr>

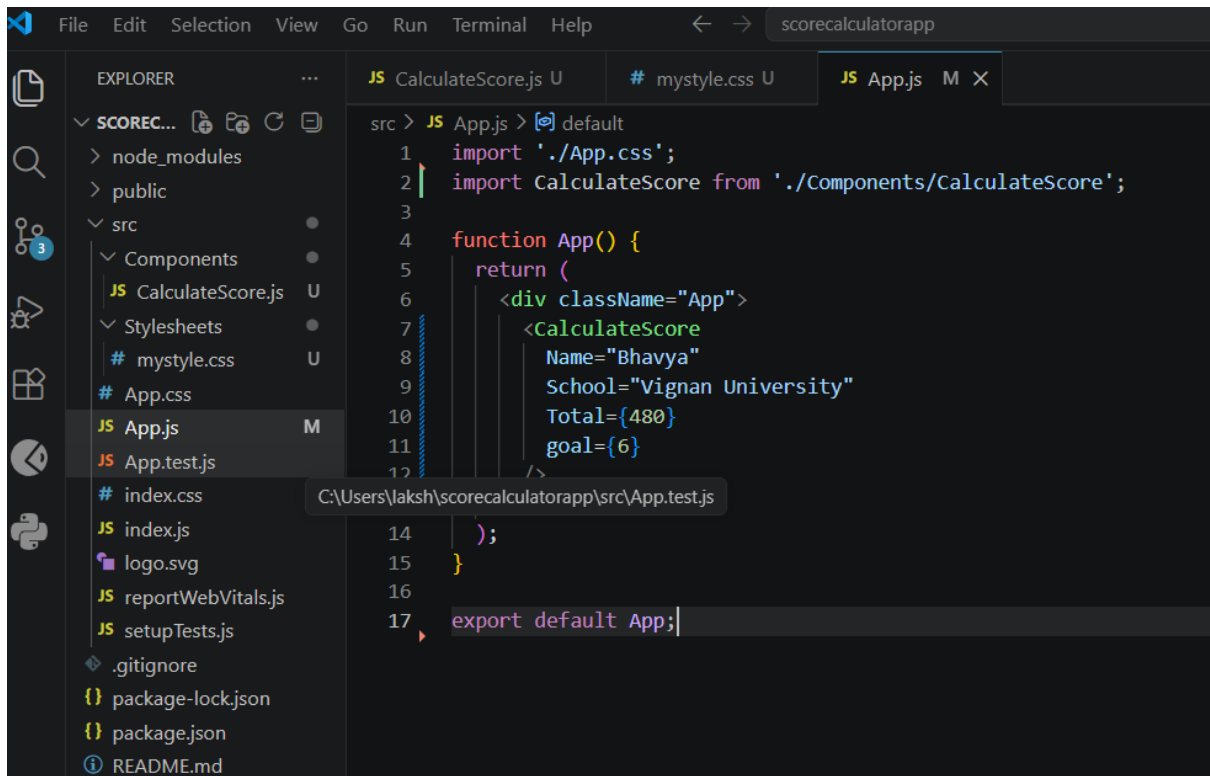
          <tr>
            <td>School</td>
            <td>{props.School}</td>
          </tr>

          <tr>
            <td>Total Marks</td>
            <td>{props.Total}</td>
          </tr>

          <tr>
            <td>Subjects</td>
            <td>{props.goal}</td>
          </tr>

          <tr>
            <td><b>Average Score</b></td>
            <td><b>{average}</b></td>
          </tr>
        </tbody>
      </table>
    </div>
  );
}
```

App.js:

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project structure with folders 'node_modules', 'public', and 'src'. Inside 'src', there are subfolders 'Components' and 'Stylesheets'. Files listed include 'CalculateScore.js', 'mystyle.css', 'App.css', 'App.js' (selected), 'App.test.js', 'index.css', 'index.js', 'logo.svg', 'reportWebVitals.js', 'setupTests.js', '.gitignore', 'package-lock.json', 'package.json', and 'README.md'. The main editor area shows the code for 'App.js'. It starts with an import for './App.css' and another for 'CalculateScore' from './Components/CalculateScore'. A function 'App()' is defined, which returns a JSX element. This element consists of a 'div' with className 'App' containing a 'CalculateScore' component. The 'CalculateScore' component has props: 'Name="Bhavya"', 'School="Vignan University"', 'Total={480}', and 'goal={6}'. The file ends with 'export default App;'. The status bar at the bottom indicates the file path: 'C:\Users\laksh\scorecalculatorapp\src\App.test.js'.

Style.css:

```
File Edit Selection View Go Run Terminal Help scorecalculatorapp

EXPLORER
SCORECALCULATORAPP
  > node_modules
  > public
  > src
    > Components
    JS CalculateScore.js U
    > Stylesheets
      # mystyle.css U
      # App.css
      JS App.js M
      JS App.test.js
      # index.css
      JS index.js
      logo.svg
      JS reportWebVitals.js
      JS setupTests.js
      .gitignore
      {} package-lock.json
      {} package.json
      README.md

src > Stylesheets > # mystyle.css > td
1 body{
2   background-color: #f2f2f2;
3   font-family: Arial, Helvetica, sans-serif;
4 }
5
6 .box{
7   width: 450px;
8   margin: 40px auto;
9   padding: 20px;
10  background: white;
11  border-radius: 10px;
12  box-shadow: 0 0 10px gray;
13  text-align: center;
14 }
15
16 h2{
17   color: darkblue;
18 }
19
20 table{
21   width: 100%;
22   margin-top: 0px;
23   border-collapse: collapse;
24   border: 0;
25 }
26
27 td{
28   border: 1px solid gray;
29   padding: 10px;
30 }
31
32 tr:nth-child(even){
33   background: #f9f9f9;
34 }
```

Output:

```
PS C:\Users\laksh\scorecalculatorapp> npm start
Starting the development server...
Compiled successfully!

You can now view scorecalculatorapp in the browser.

Local:      http://localhost:3000
On Your Network: http://10.180.5.36:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

